Internship report

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1 Introduction

The objective of this internship was to design and validate a greedy algorithm for refining Constrained Delaunay Tetrahedrizations (CDTs).

CDTs are commonly used in physics-based simulations such as the finite element method. However, these simulations require high-quality meshes, which raw CDTs often fail to provide. Therefore, a refinement process is essential to improve the overall mesh quality and ensure the robustness and accuracy of the simulations.

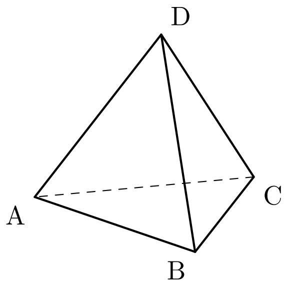
2 Preliminary

2.1 Delaunay's triangulation

2.2 Piecewise Linear Complex

2.3 Constrained Delaunay Triangulation

2.4 Quality Measure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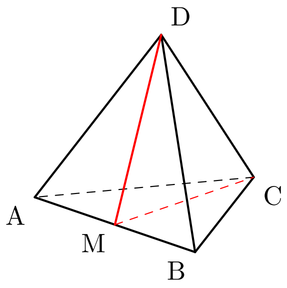
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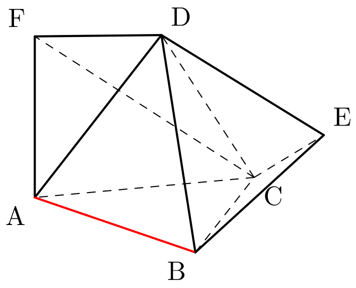
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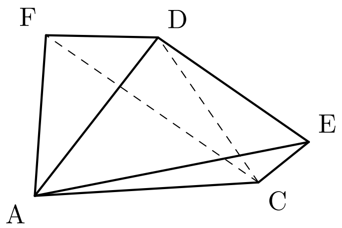
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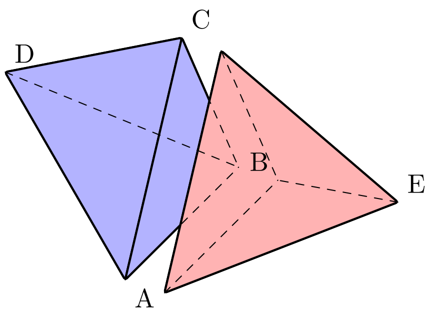
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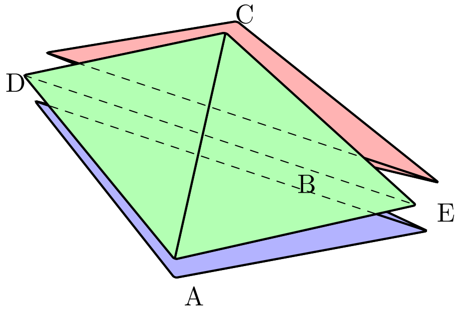
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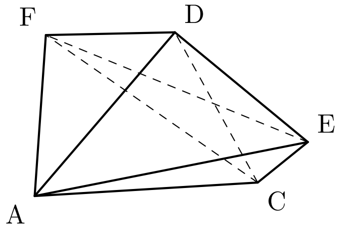
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